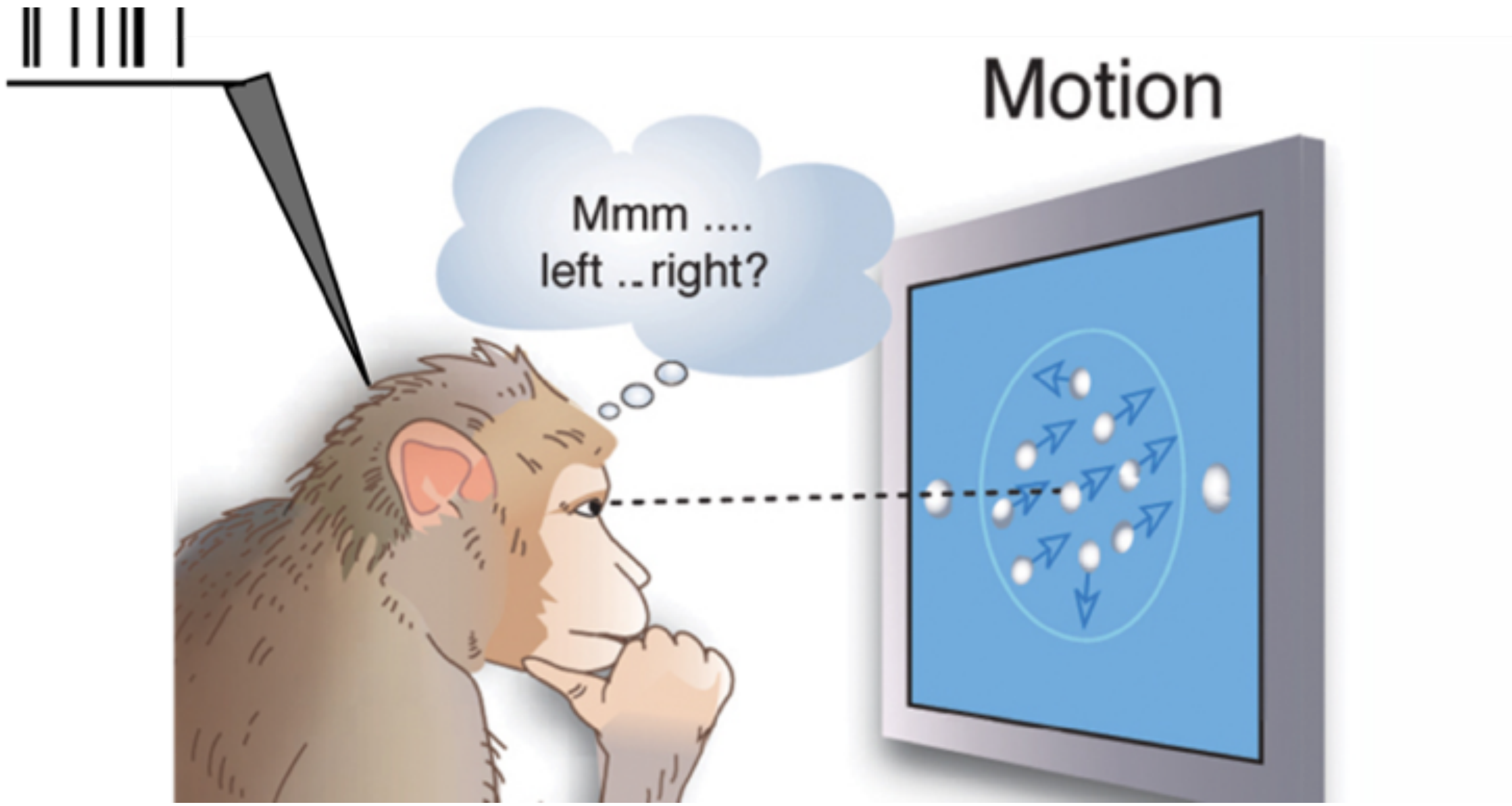
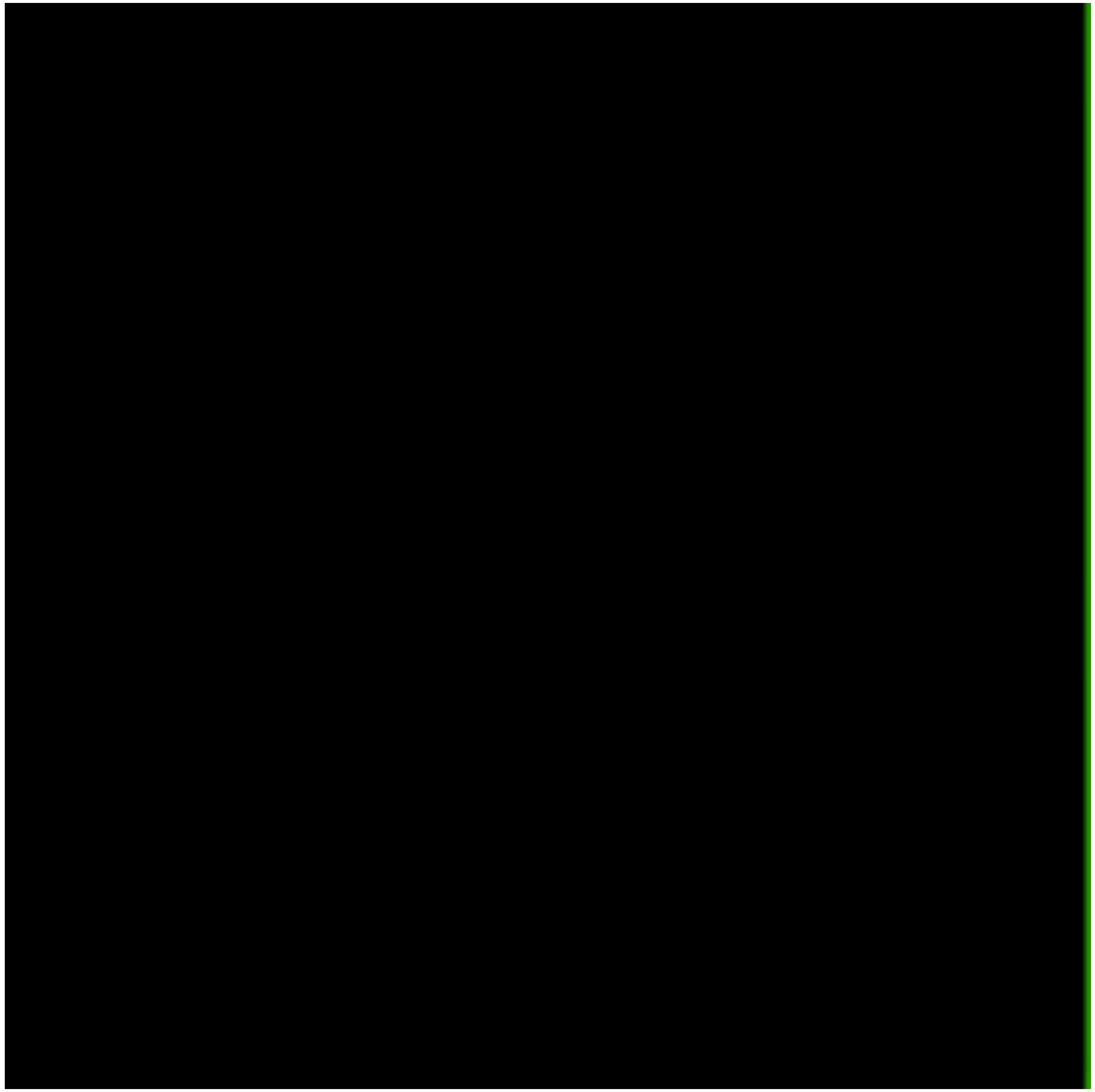
Neuron Decoding

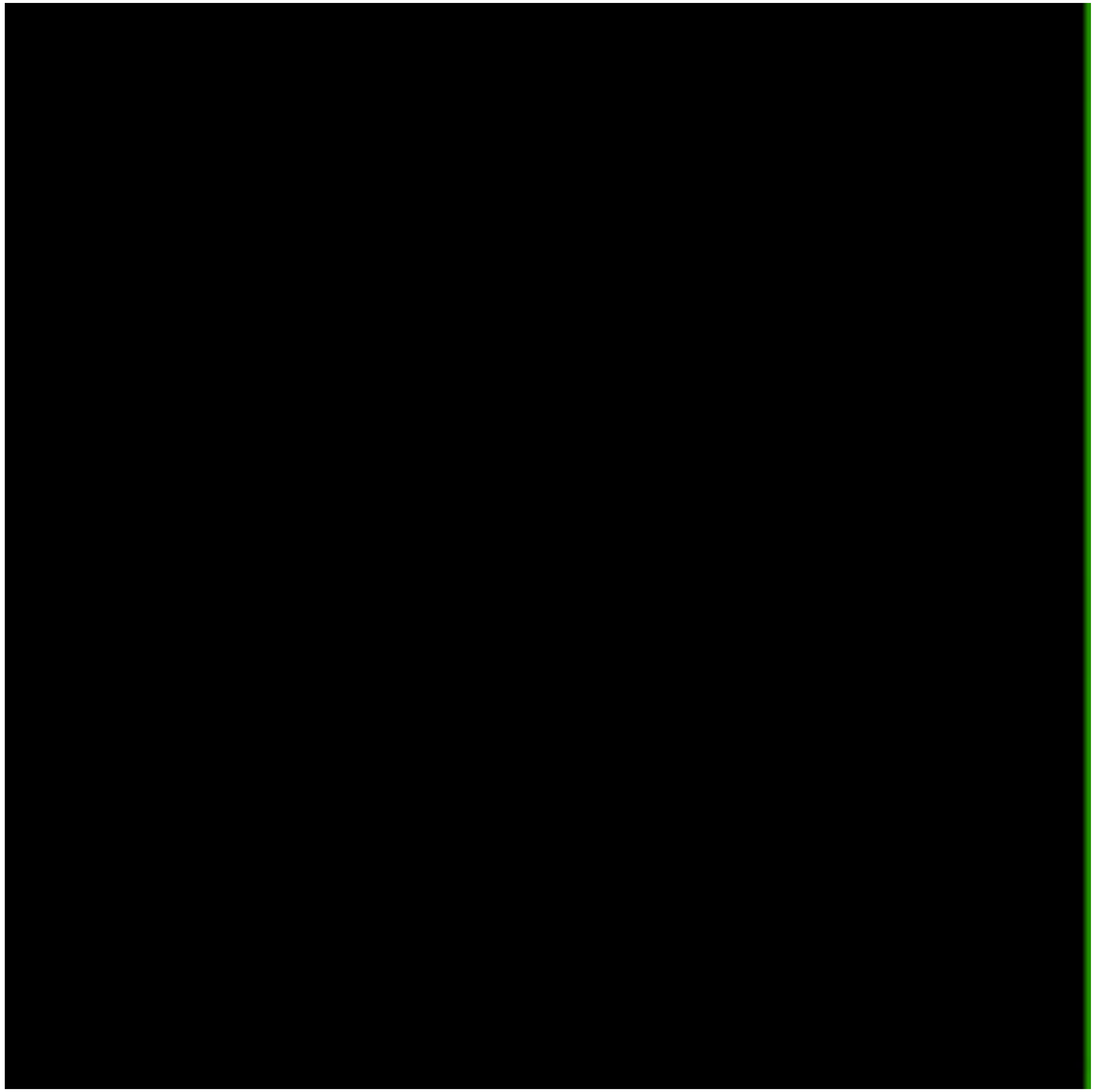
Random dots motion discrimination task



This slide and the following movies are adapted from Professor Tianming Yang's lectures



100%



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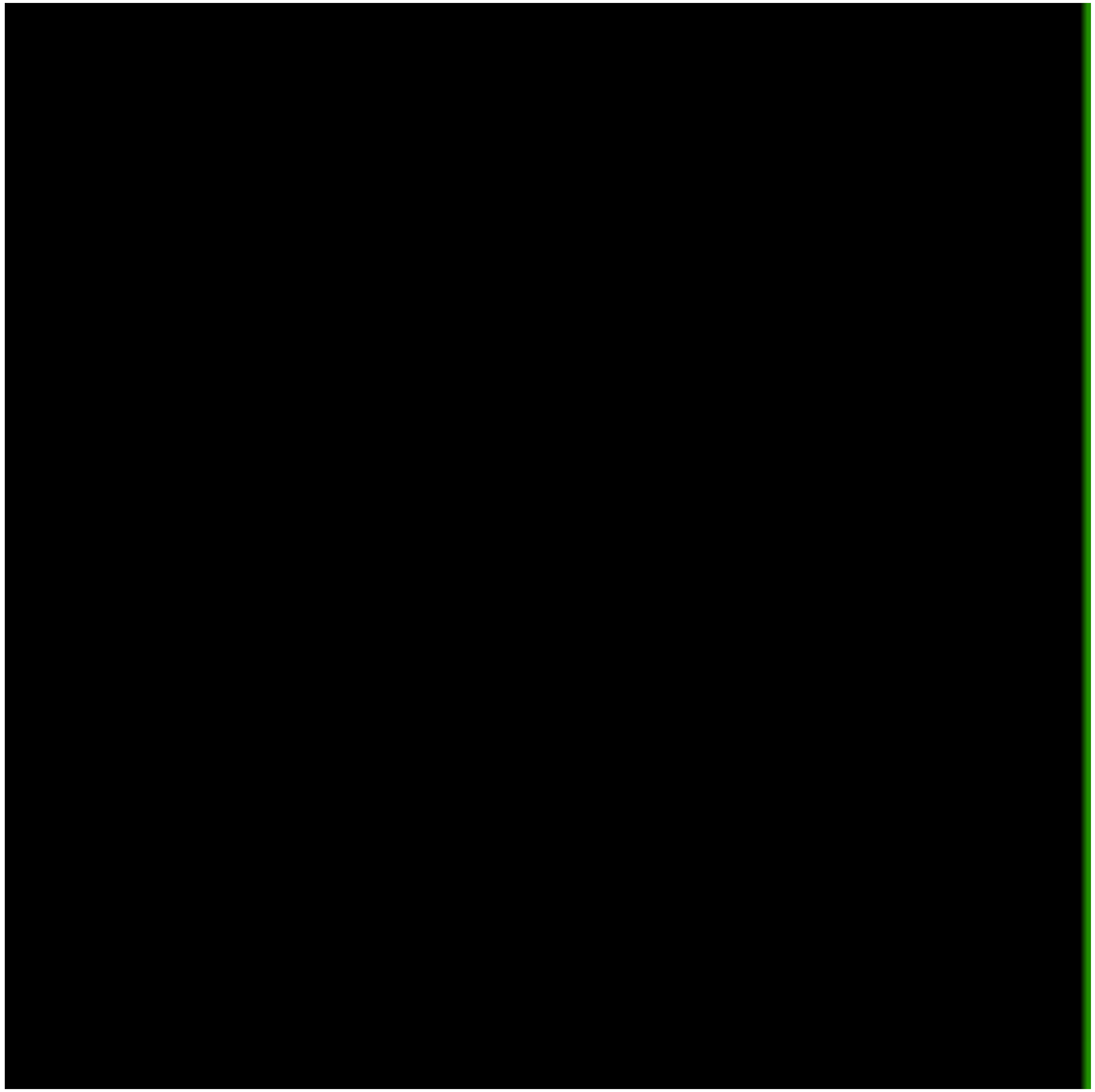

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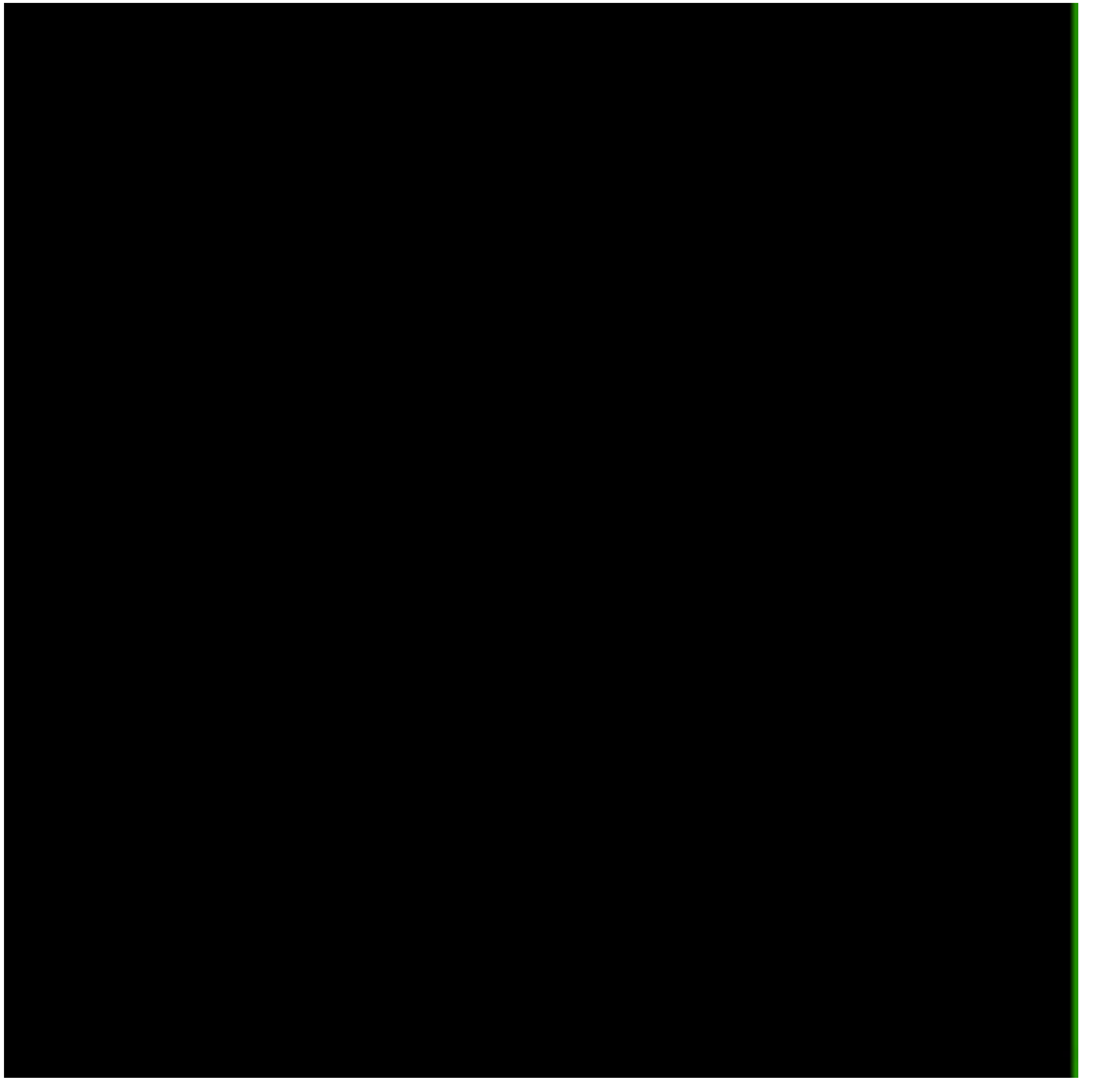

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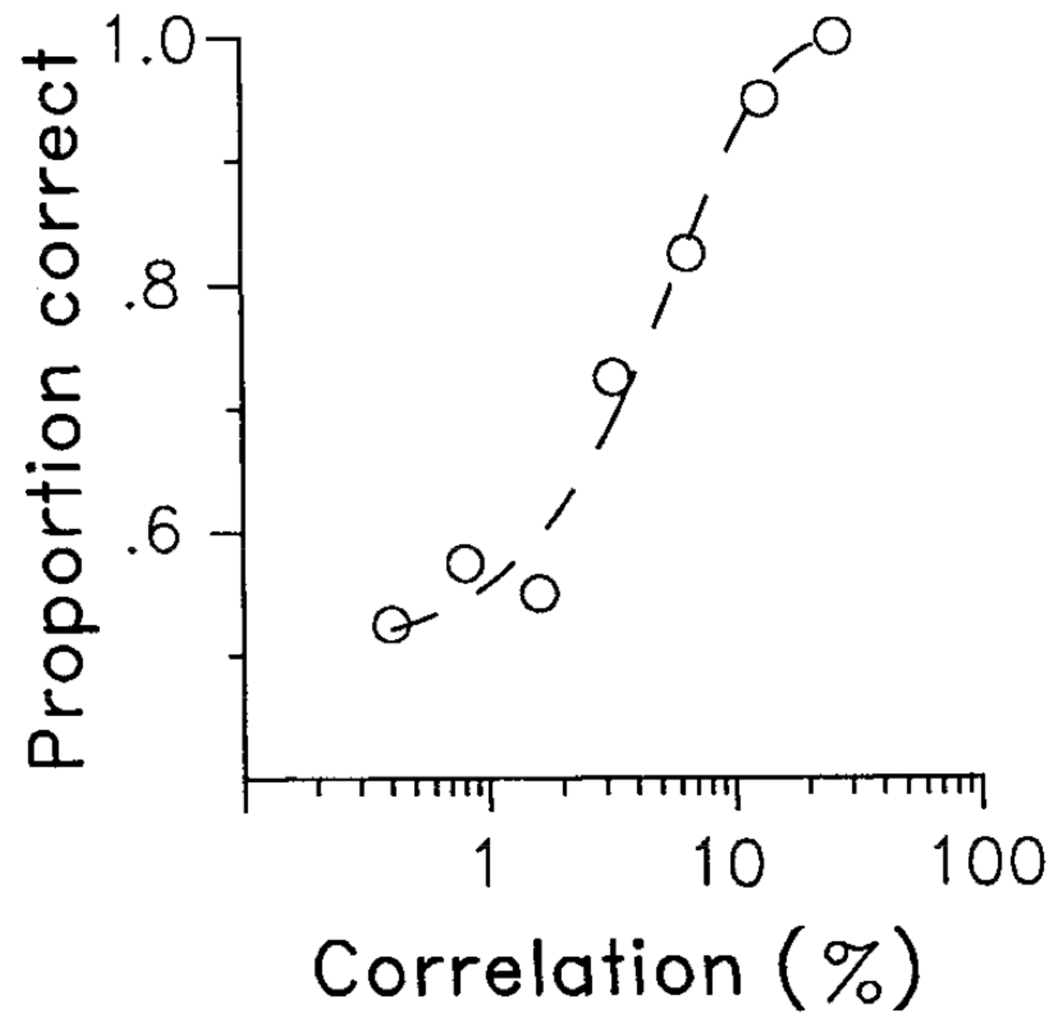
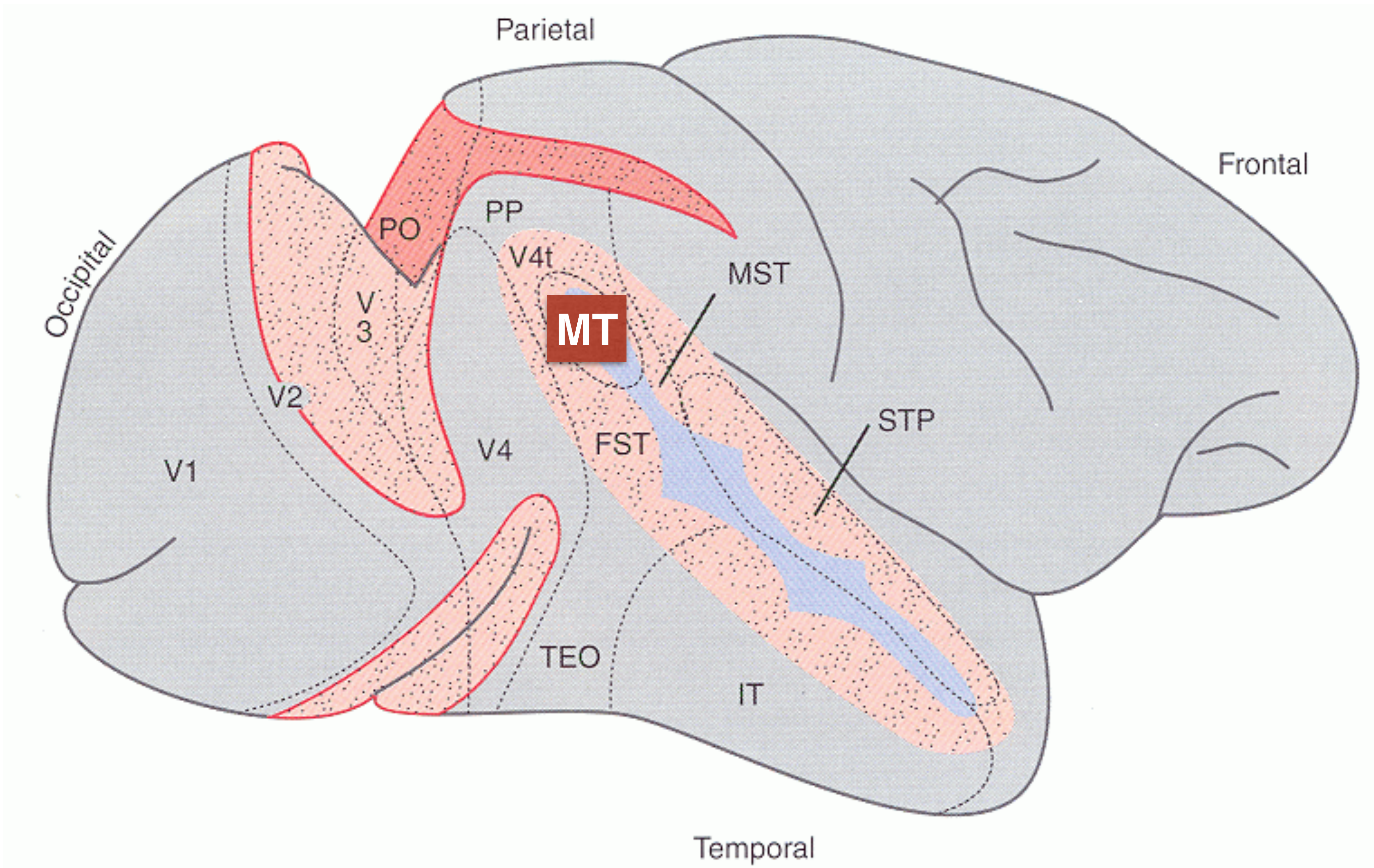
Behavior Performance



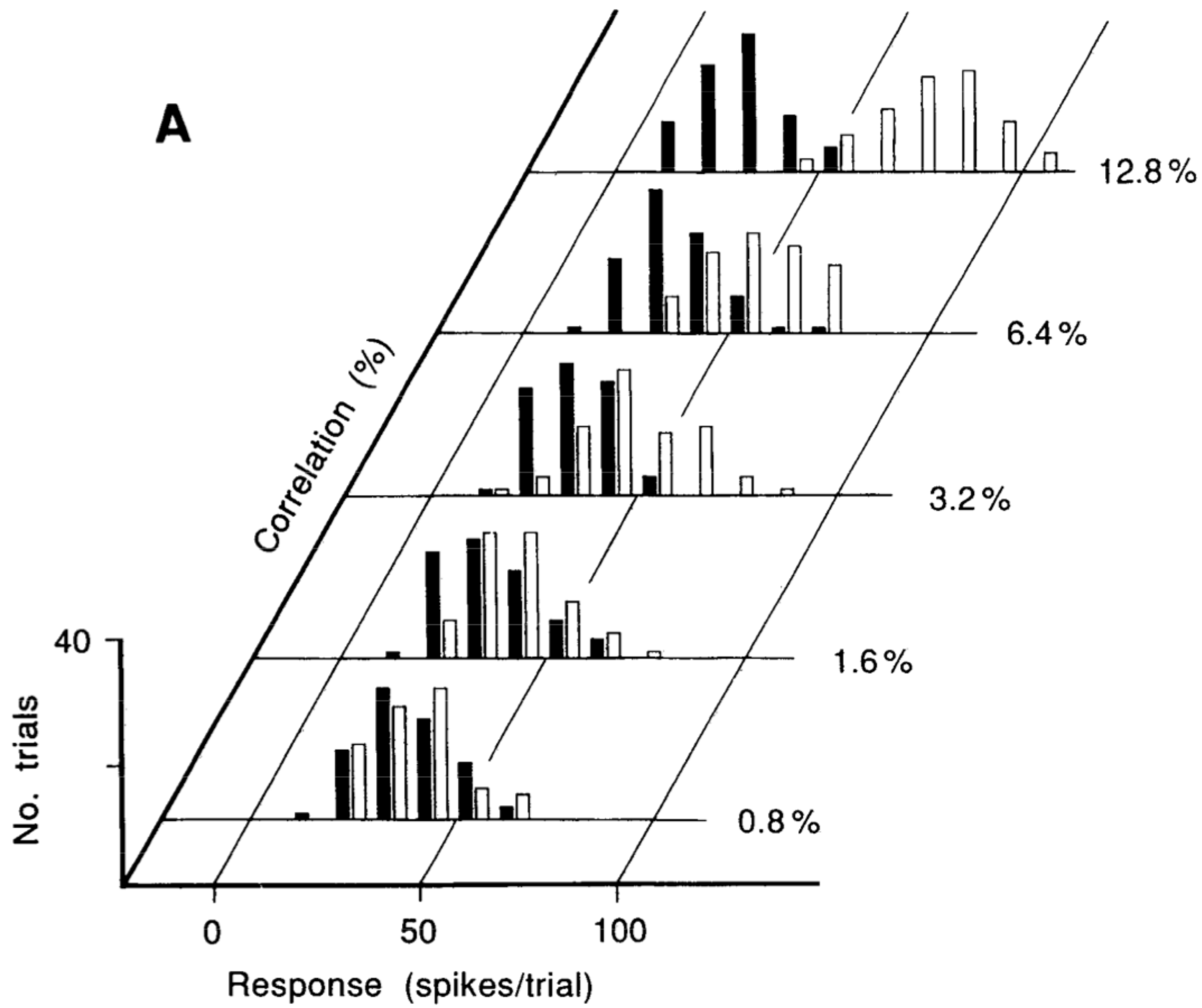
Britten, Shadlen, Newsome and Movshon, J Neurosci., 1992

Measuring neural activity in MT



http://fourier.eng.hmc.edu/e180/lectures/figures/macaquebrain.gif

Single Neuron recording



Britten, Shadlen, Newsome and Movshon, J Neurosci., 1992

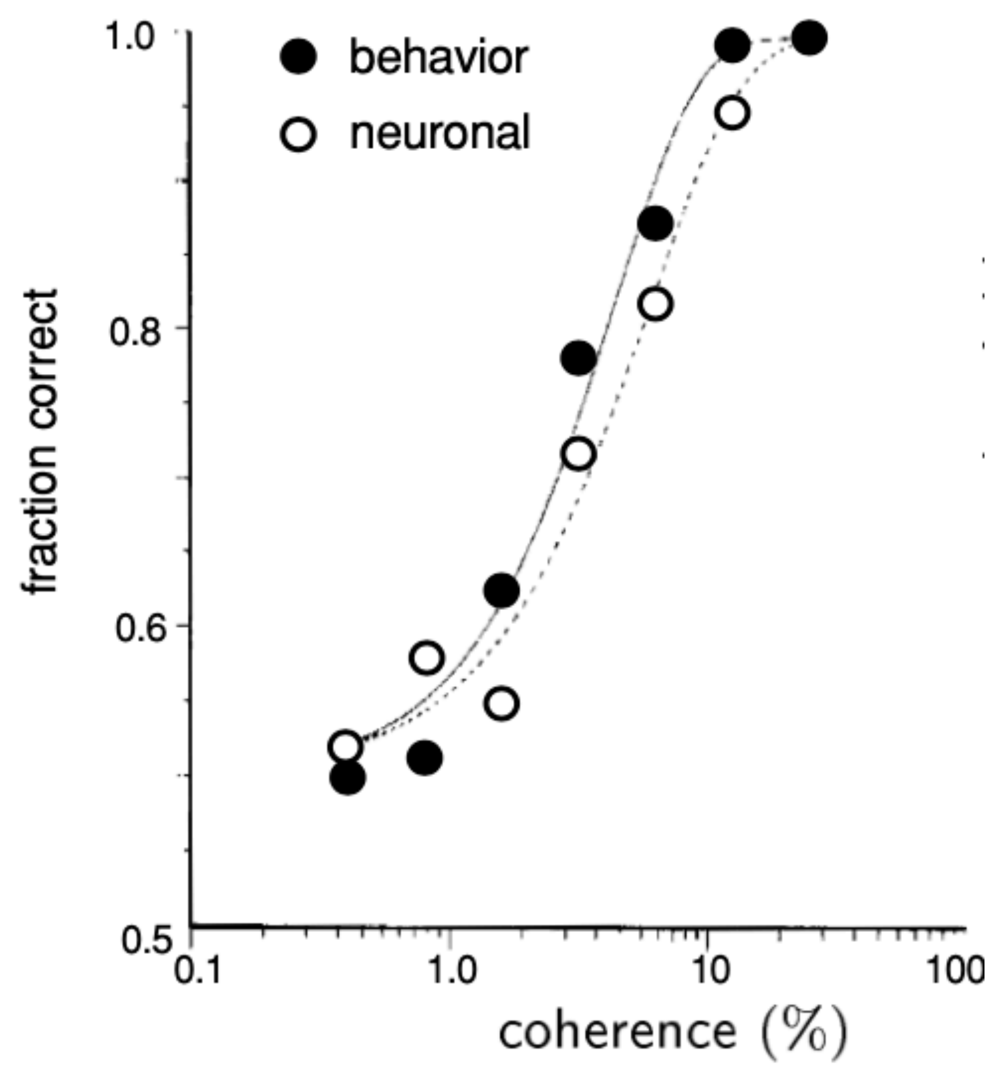
Decode stimulus direction from neural activity

	probability	
stimulus	correct	incorrect
$+$	β	$1-\beta$
$-$	$1-\alpha$	α

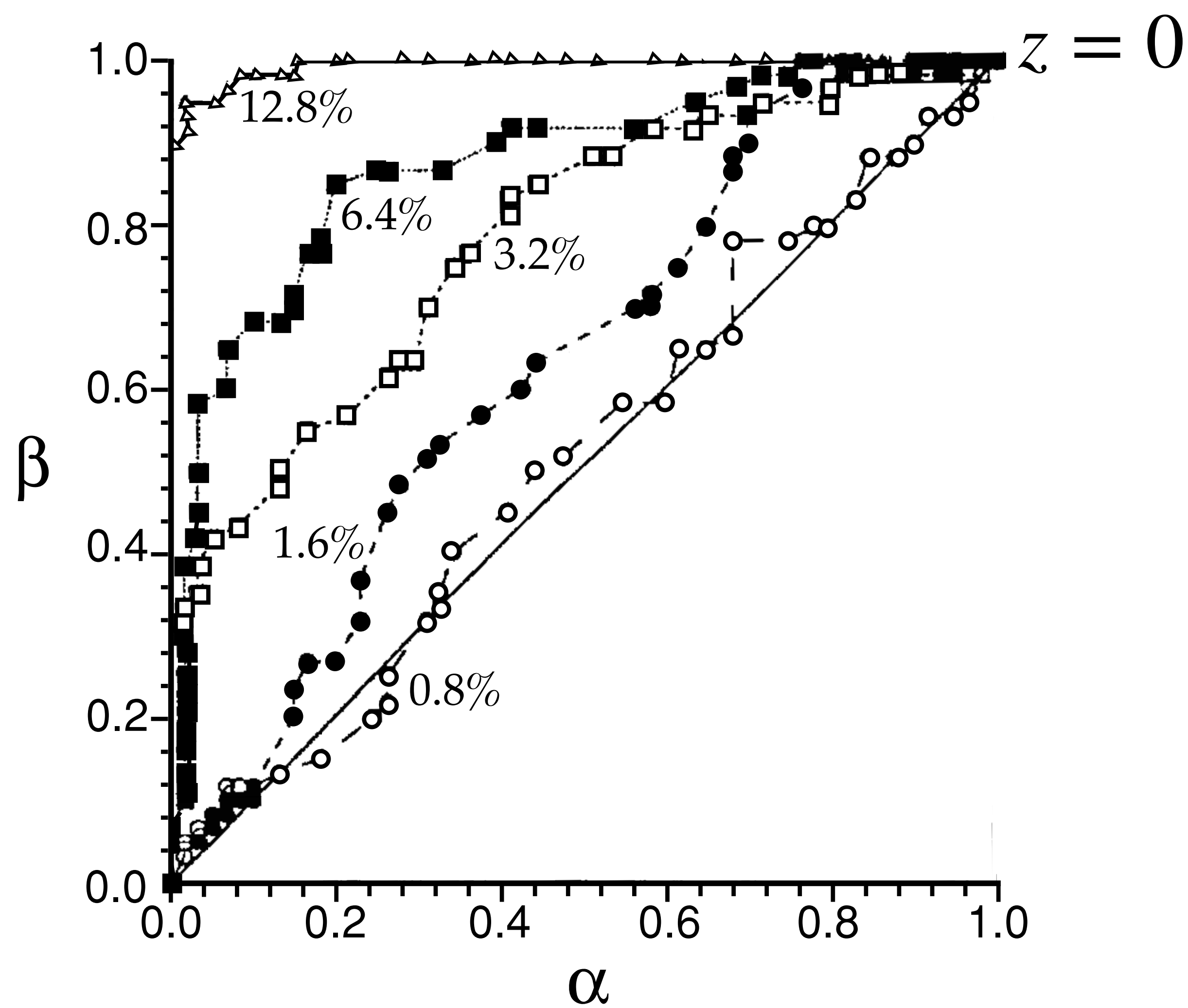
$\alpha(z) = P[r \geq z \mid -]$ false alarm rate

$\beta(z) = P[r \geq z \mid +]$ hit rate

Neural Correlates with Behavior



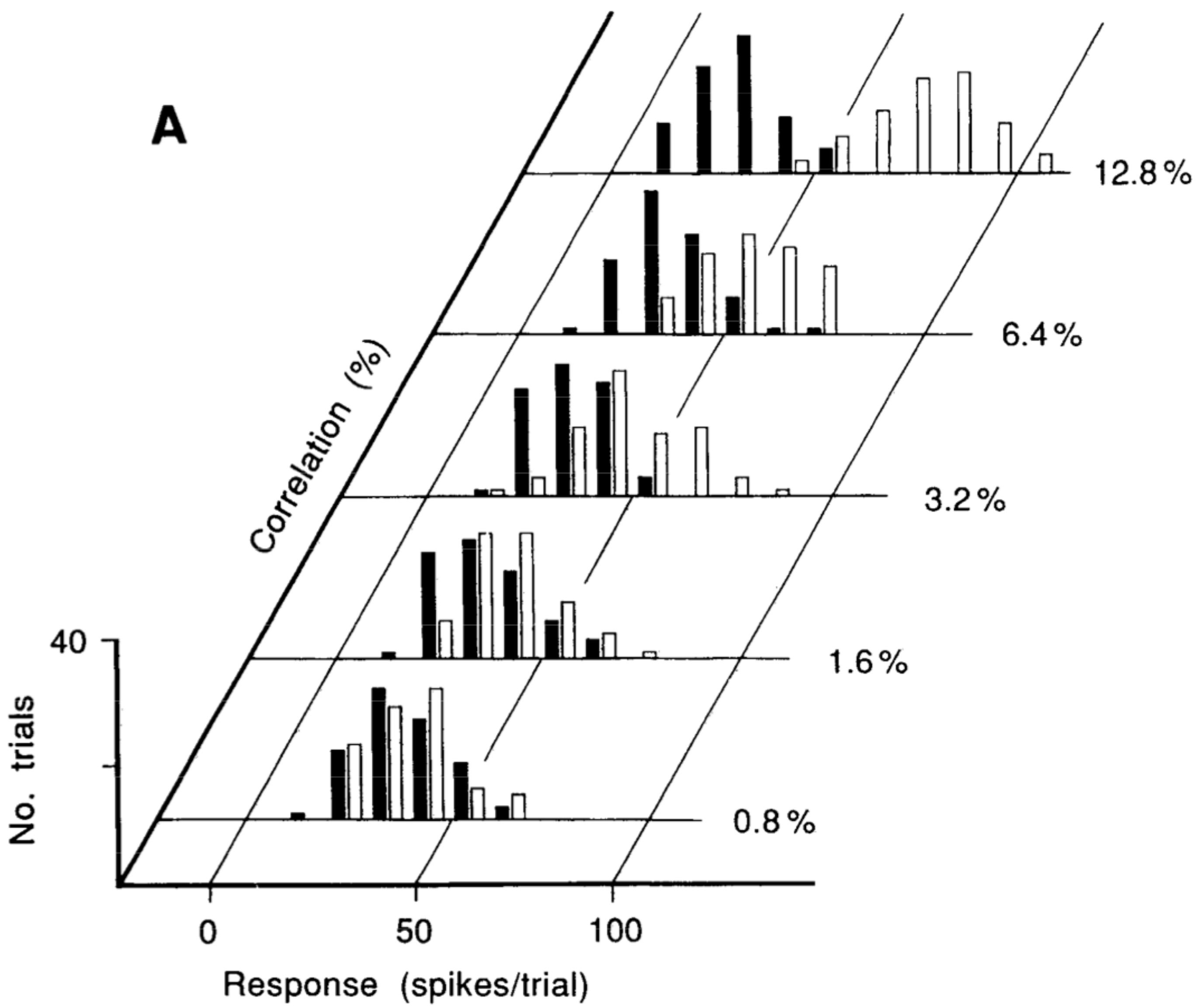
ROC (receiver operating characteristic) curve



$\alpha(z) = P[r \geq z | -]$ false alarm rate

$\beta(z) = P[r \geq z | +]$ hit rate

Discriminability (or SNR)



$$d = \frac{\langle r \rangle_+ - \langle r \rangle_-}{\sigma}$$

$$P(\text{correct}) = \frac{1}{\sqrt{\pi}} \int_{-d/2}^{\infty} dx \exp(-x^2)$$

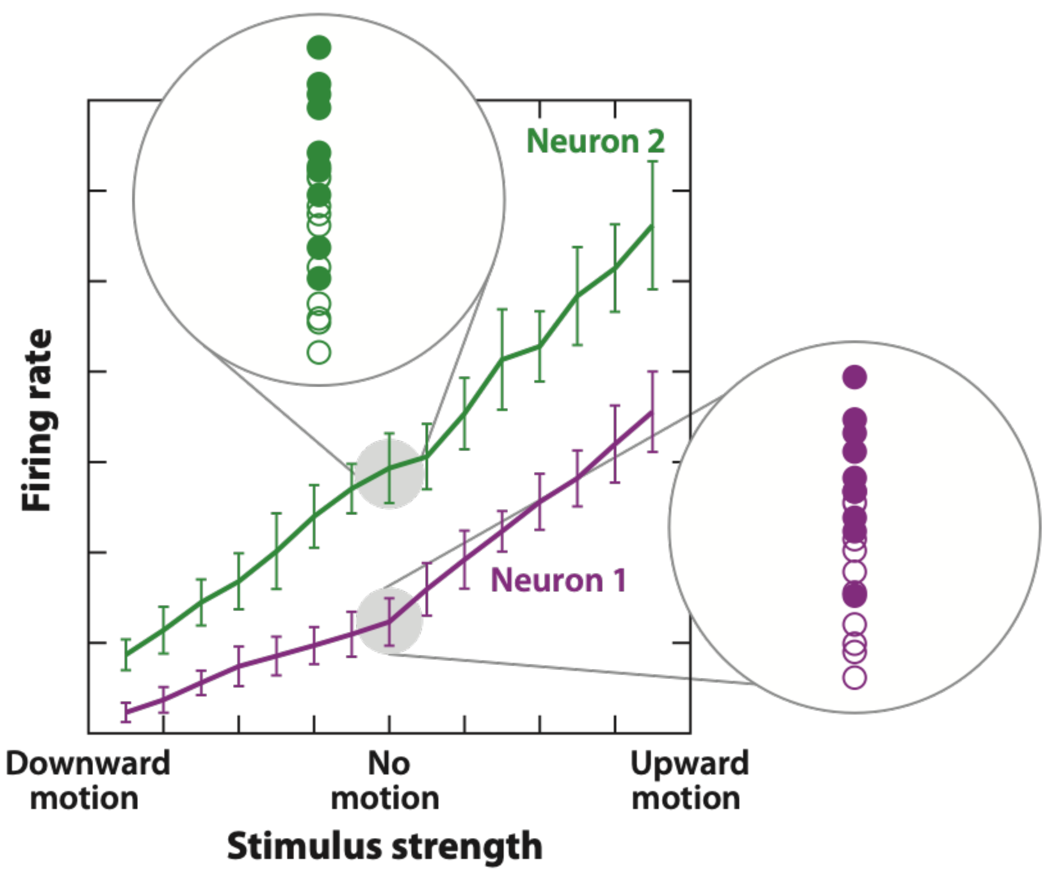
Britten, Shadlen, Newsome and Movshon, J Neurosci., 1992

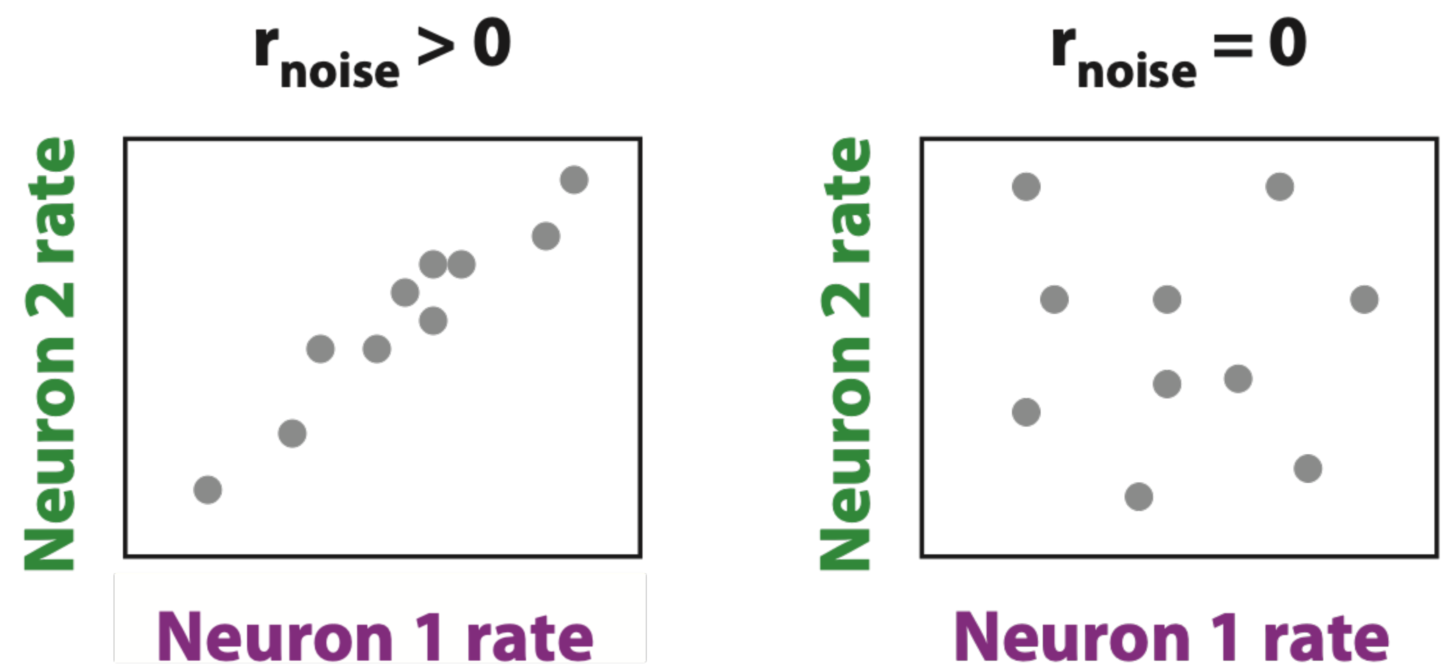
Discrimination (decoding) with population of neurons

$$k_+ = \sum_{i=1}^{N} r_+^i \qquad k_- = \sum_{i=1}^{N} r_-^i$$

$$d^2 = \frac{\left(\langle k_+ \rangle - \langle k_- \rangle\right)^2}{\frac{1}{2}\langle\left(k_+ - \langle k_+ \rangle\right)^2\rangle + \frac{1}{2}\langle\left(k_- - \langle k_- \rangle\right)^2\rangle}$$

Decoding with correlated neurons





Signal Correlation

Noise Correlation

Decoding with correlated neurons (a classic argument)

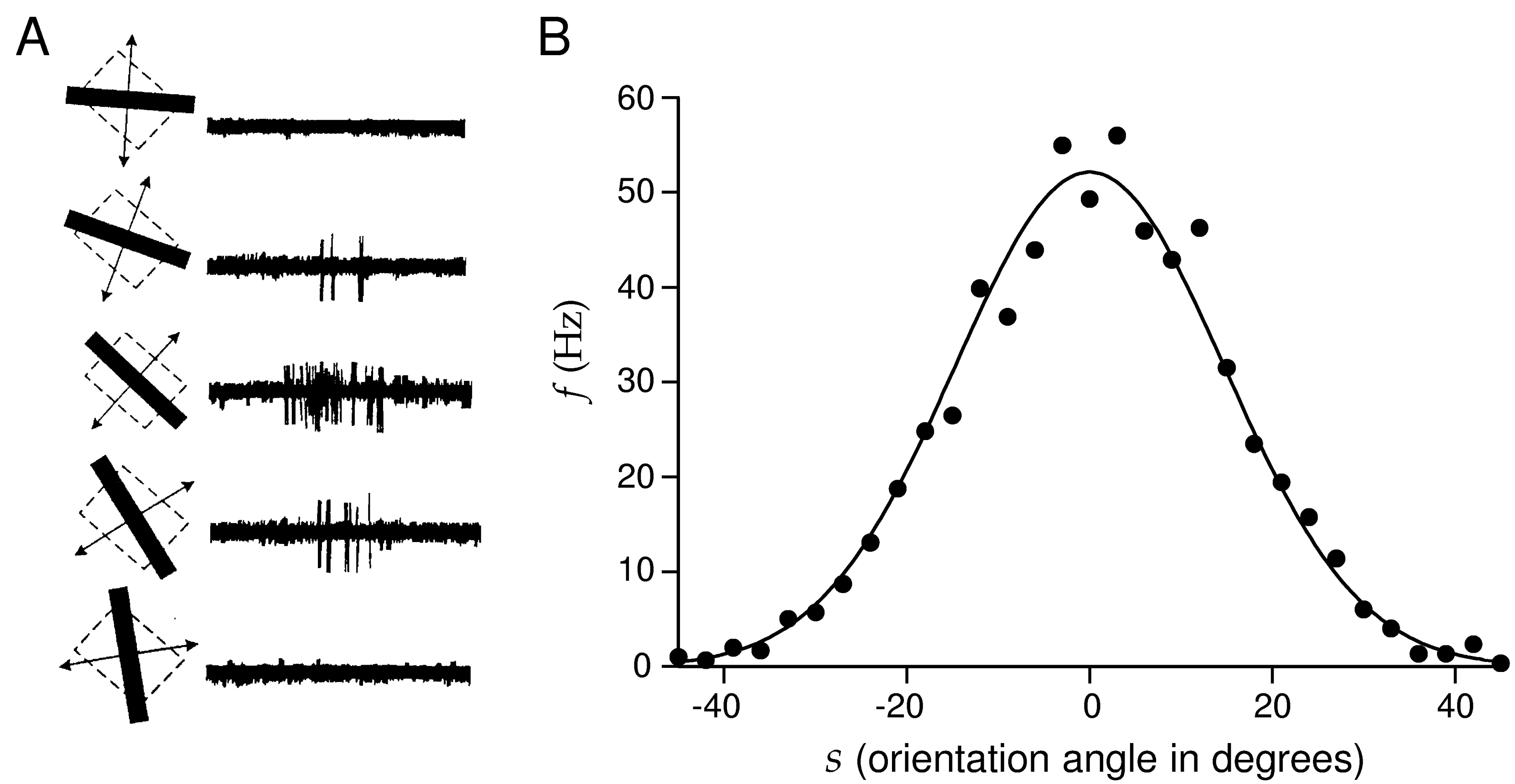
$$k_+ = \sum_{i=1}^{N} r_+^i \qquad k_- = \sum_{i=1}^{N} r_-^i$$

$$d^2 = \frac{\big(\langle k_+\rangle - \langle k_-\rangle\big)^2}{\frac{1}{2}\langle\big(k_+ - \langle k_+\rangle\big)^2\rangle + \frac{1}{2}\langle\big(k_- - \langle k_-\rangle\big)^2\rangle}$$

$$\big(\langle k_+\rangle - \langle k_-\rangle\big)^2 \sim N^2\langle r_+ - r_-\rangle^2$$

$$\langle\big(k_+ - \langle k_+\rangle\big)^2\rangle = N\langle \delta r_i^2\rangle + N(N-1)\langle \delta r_i \delta r_j\rangle$$

Orientation selectivity in the visual cortex and fine discrimination



Gaussian tuning curve of a neuron in the primary visual cortex (V1)

What is the optimal linear decoder?

$$s_+ = \overrightarrow{w} \cdot \overrightarrow{r_+} \qquad s_- = \overrightarrow{w} \cdot \overrightarrow{r_-}$$

$$d^2 = \frac{(\langle v_+ \rangle - \langle v_- \rangle)^2}{\frac{1}{2}(\delta v_+^2 + \delta v_-^2)}$$

What is the optimal linear decoder?

$s_+ = \vec{w} \cdot \vec{r_+}$ $\qquad$ $s_- = \vec{w} \cdot \vec{r_-}$

$$d^2 = \frac{(\langle v_+ \rangle - \langle v_- \rangle)^2}{\frac{1}{2}(\delta v_+^2 + \delta v_-^2)}$$

$$\overrightarrow{w_{opt}} = \Sigma^{-1}\big(\langle \vec{r_+} \rangle - \langle \vec{r_-} \rangle\big) = \Sigma^{-1}\Delta\mu$$

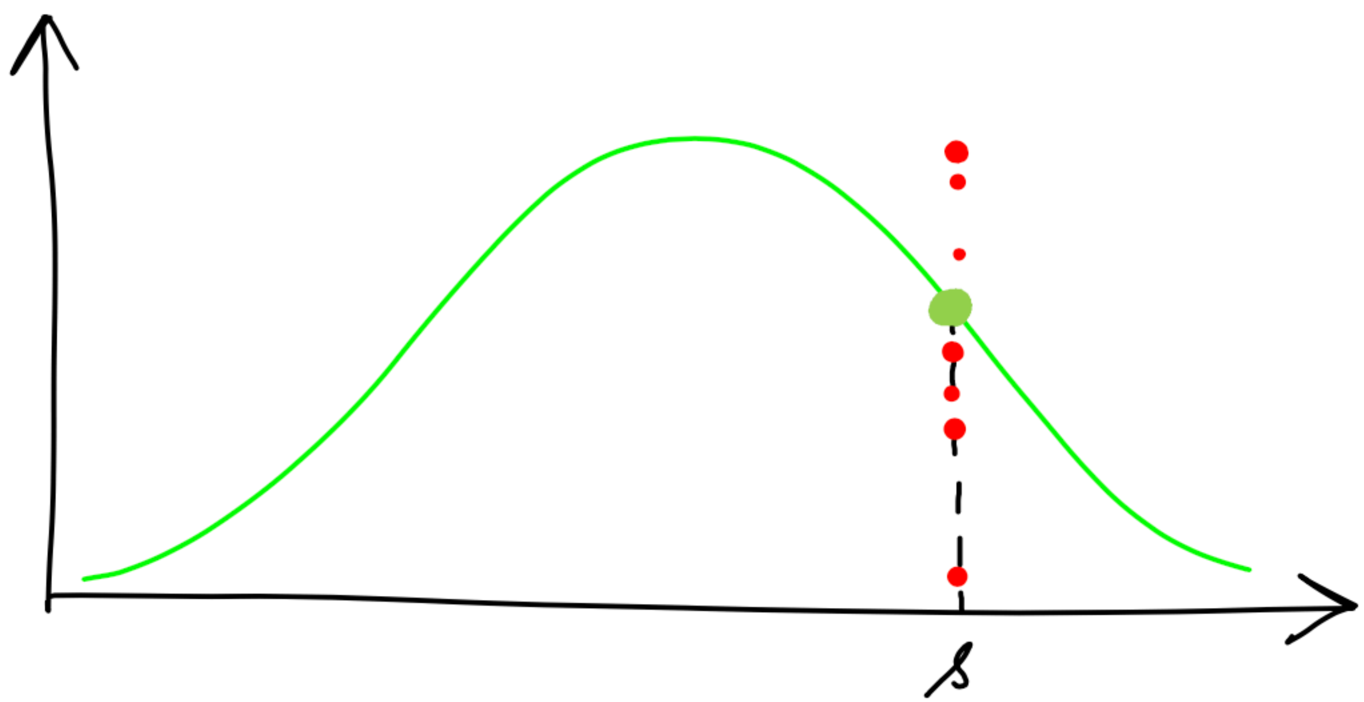
$$d^2 = \Delta\mu^T \Sigma^{-1} \Delta\mu$$

$$\Sigma_{ij} = \langle \delta r_i \delta r_j \rangle$$

Bayesian decoding method

Variability

$$r = m(s) + \xi$$



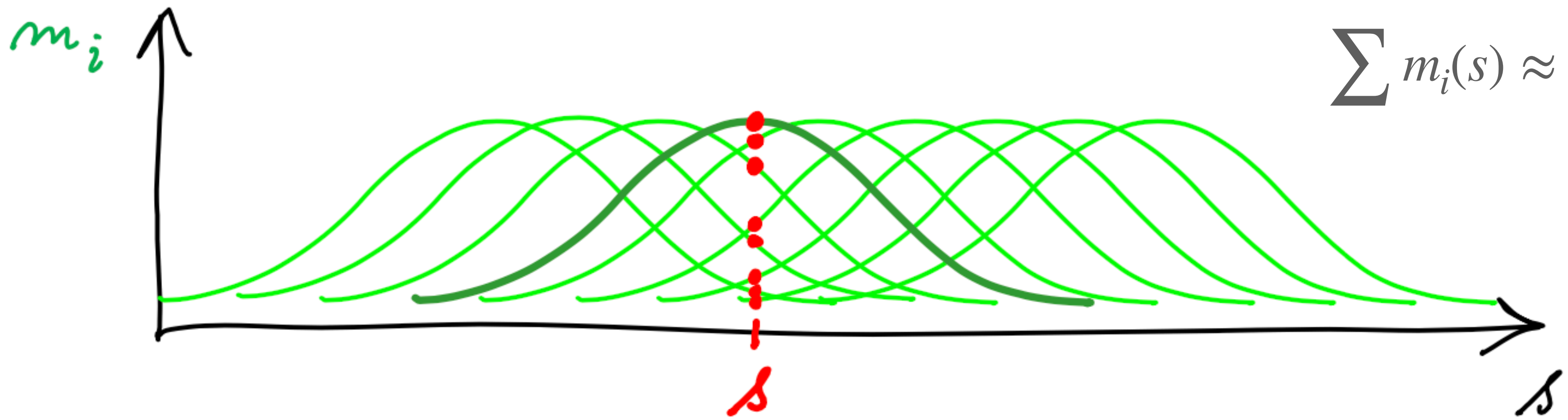
Likelihood Prior

$$p(s\,|\,r) = \frac{p(r\,|\,s)p(s)}{p(r)}$$

Normalization factor

$$m(s) = r_{max}\exp\left(-\frac{1}{2}(s - s_i)^2/\sigma_i^2 \right)$$

Population decoding



$\sum m_i(s) \approx$ const

$$p(n_i \mid s) = \frac{[m_i(s)T]^{n_i}}{n_i!} \exp(-m_i(s)T)$$

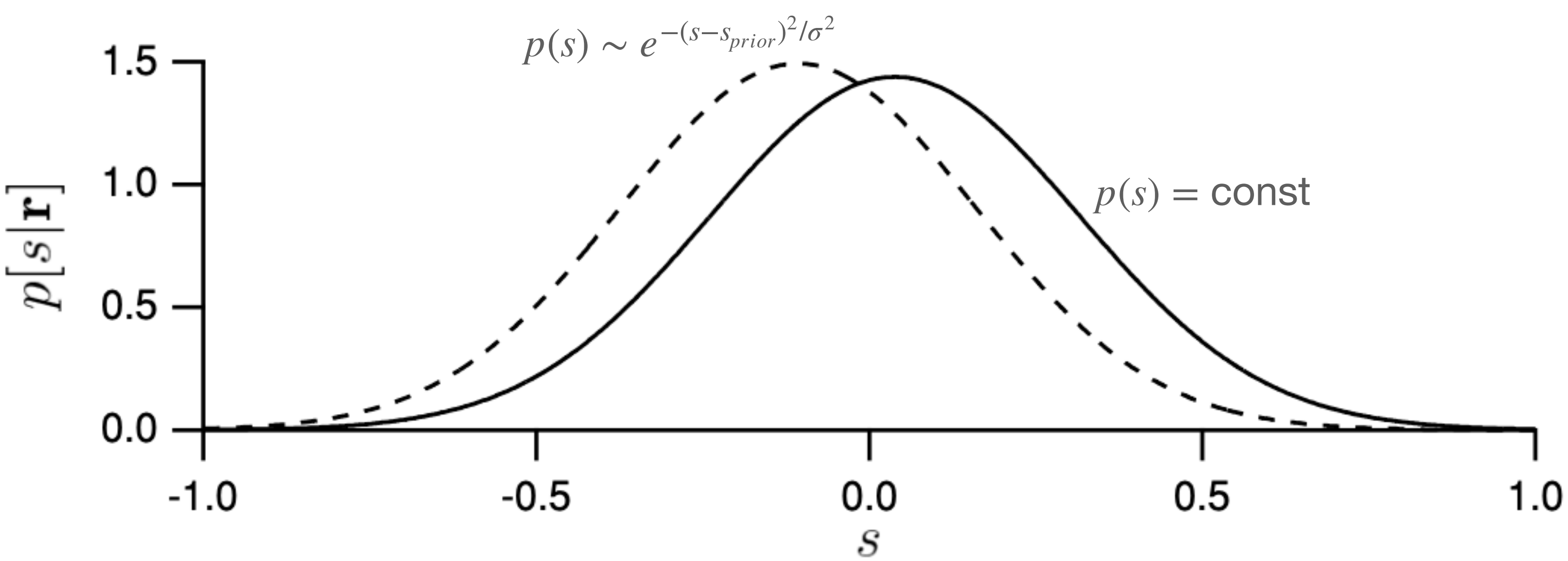
$$\operatorname*{argmax}_s \log p(\mathbf{r} \mid s) = \sum_i \log p(n_i \mid s)$$

$$m_i'(s)/m_i(s) = (s_i - s)/\sigma_i^2$$

$$\sum_i^N r_i \frac{m_i'(s)}{m_i(s)} = 0$$

$$s_{est} = \frac{\sum r_i s_i/\sigma_i^2}{\sum r_i/\sigma_i^2}$$

How good is our decoding method?



$$s - \langle s_{est} \rangle = b_{est}$$ bias

$$(\langle s_{est} \rangle - s_{est})^2 = \sigma_{est}^2$$ Estimated variance

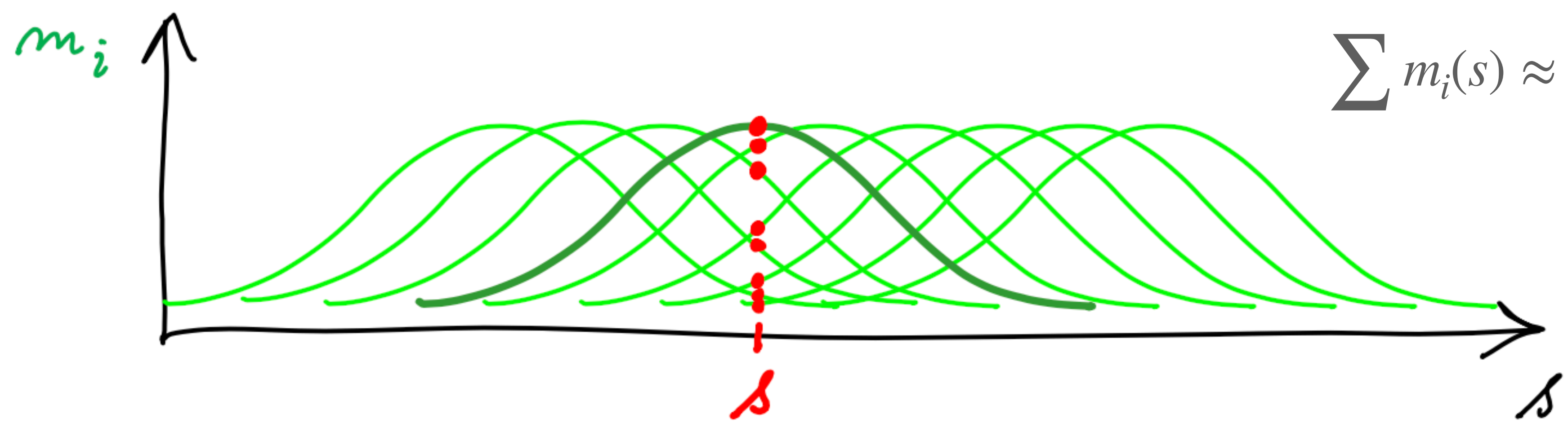
$$\langle (s - s_{est})^2 \rangle = \langle (s - \langle s_{est} \rangle + \langle s_{est} \rangle - s_{est})^2 \rangle = b_{est}^2 + \sigma_{est}^2$$ Total Variance

Cramer-Rao bound and Fisher information

$$\sigma_{est}^2 \geq \frac{(1 + b'_{est})^2}{I_F}$$

$$I_F = \left\langle -\frac{\partial^2 \log p(r|s)}{\partial s^2} \right\rangle = -\int dr p(r|s) \frac{\partial^2 \log p(r|s)}{\partial s^2}$$

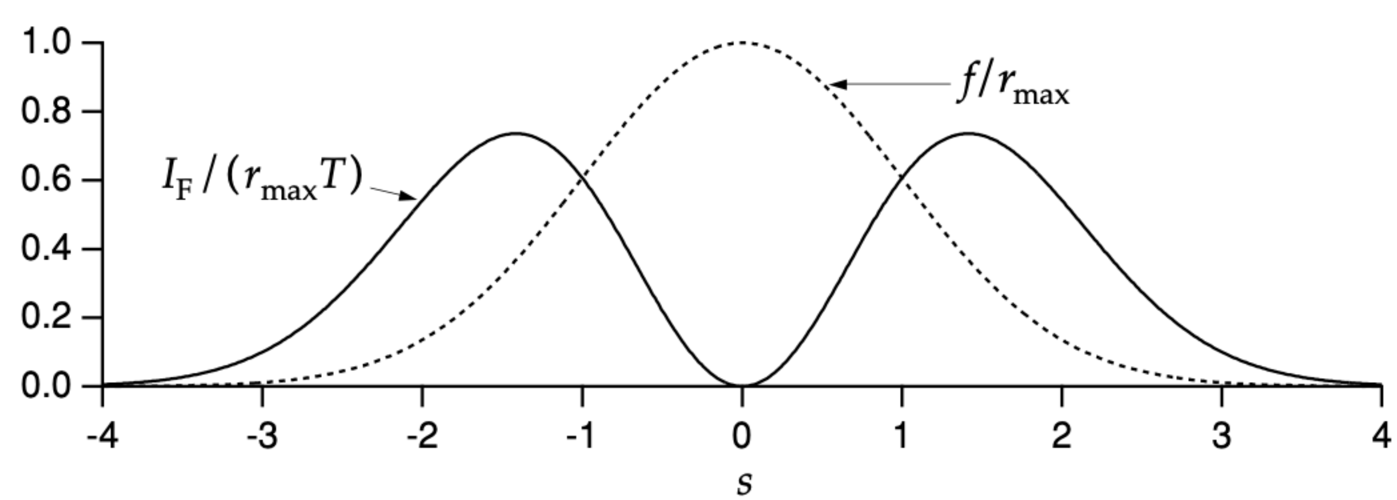
Population decoding and fisher information



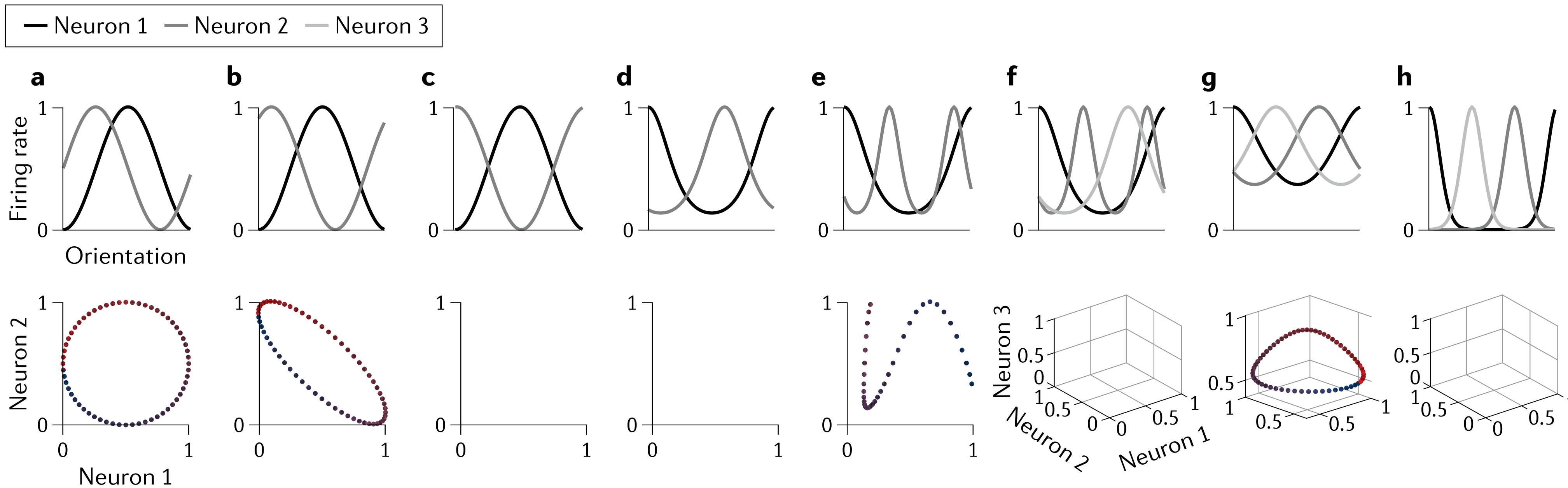
$\sum m_i(s) \approx$ const

$$I_F = T \sum_i \frac{\left(m_i'(s)\right)^2}{m_i(s)}$$

independent poisson neurons



Neuron tuning and representation geometry

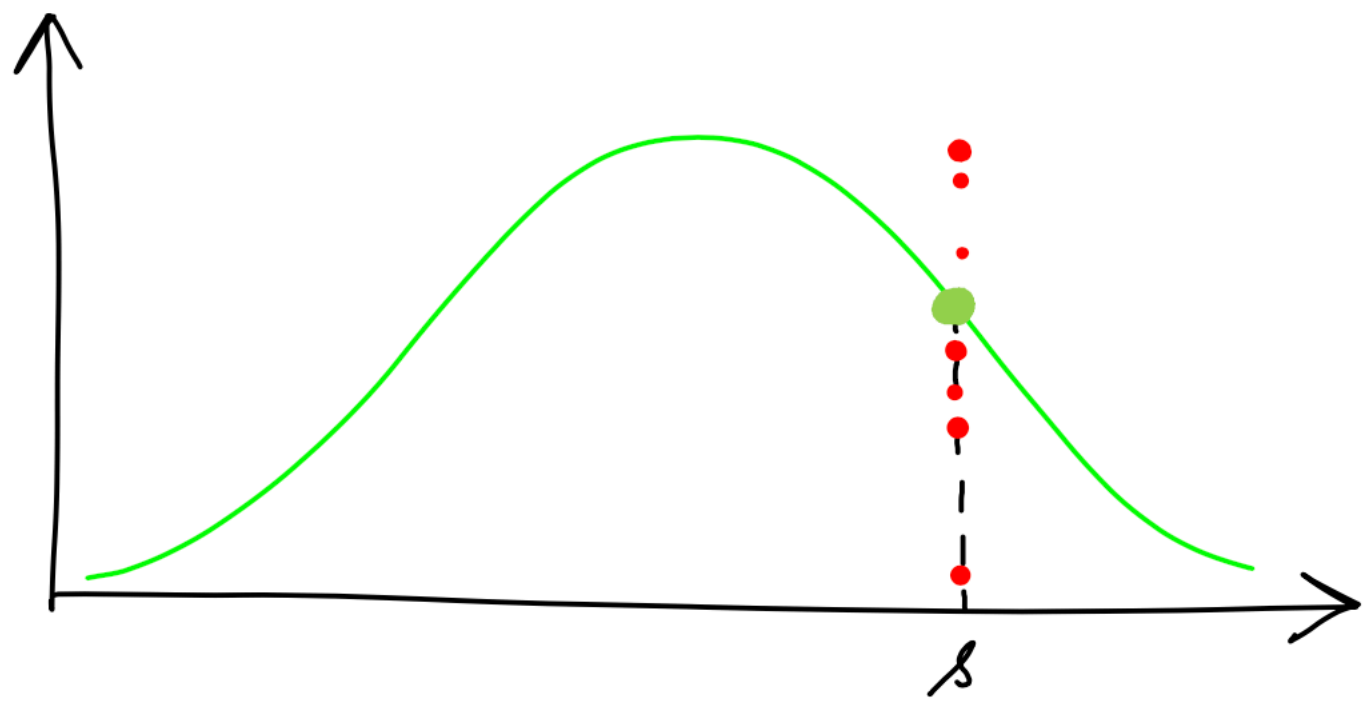


Kriegeskorte and Wei 2021

Fisher information of Gaussian ensemble

Variability

$$r = m(s) + \xi$$



$$p(\mathbf{r}\,|\,s) \sim \frac{1}{Z}\exp\left[-\frac{1}{2}(\mathbf{r}-\mathbf{m}(s))^T\boldsymbol{\Sigma}^{-\mathbf{1}}(\mathbf{r}-\mathbf{m}(\mathbf{s}))\right]$$

$$I_F = \mathbf{m}'(s)^T\boldsymbol{\Sigma}^{-1}\mathbf{m}'(s)$$

What is the optimal linear decoder?

$$v_A = \overrightarrow{w} \cdot \overrightarrow{r_A} \qquad\qquad v_B = \overrightarrow{w} \cdot \overrightarrow{r_B}$$

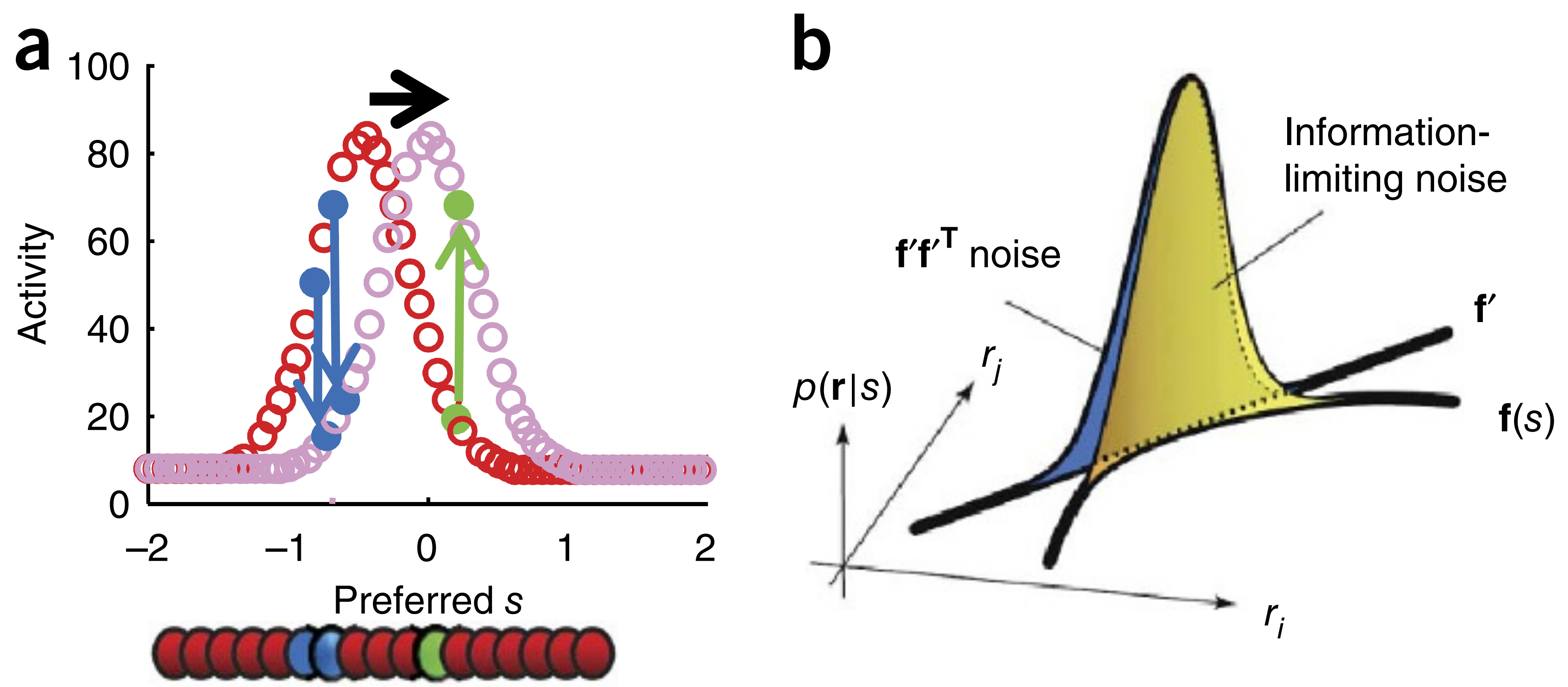
$$d^2 = \frac{(\langle v_A \rangle - \langle v_B \rangle)^2}{\delta v_A^2 + \delta v_B^2}$$

$$\overrightarrow{w_{opt}} = \Sigma^{-1}\left(\langle \overrightarrow{r_A} \rangle - \langle \overrightarrow{r_B} \rangle\right) = \Sigma^{-1} \Delta\mu$$

$$d^2 = \Delta\mu^T \Sigma^{-1} \Delta\mu$$

$$\Sigma_{ij} = \langle \delta r_i \delta r_j \rangle$$

Differential Correlation limits information coding



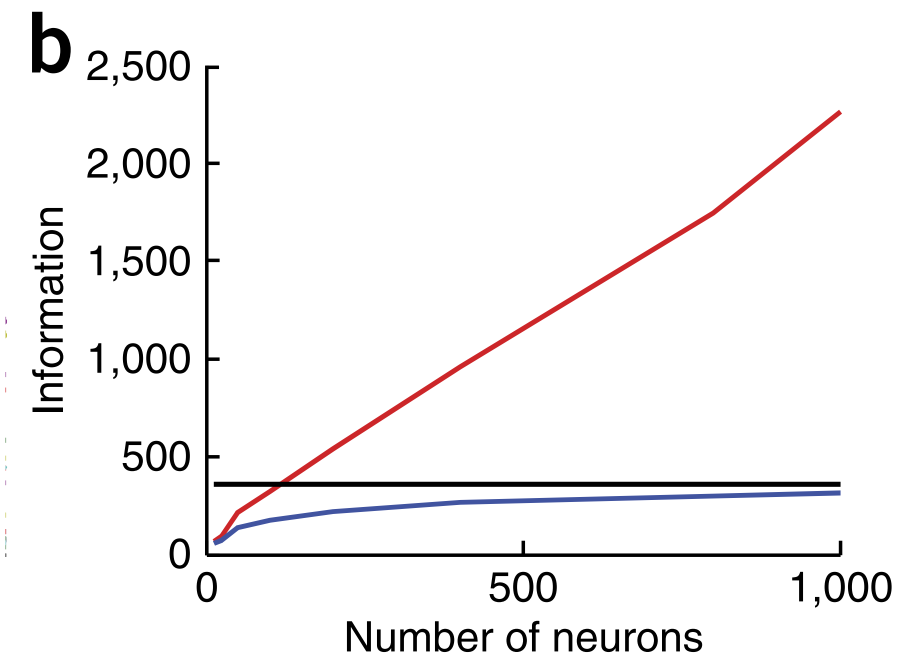
Rubén Moreno-Bote et al, 2014

Differential Correlation limits information coding

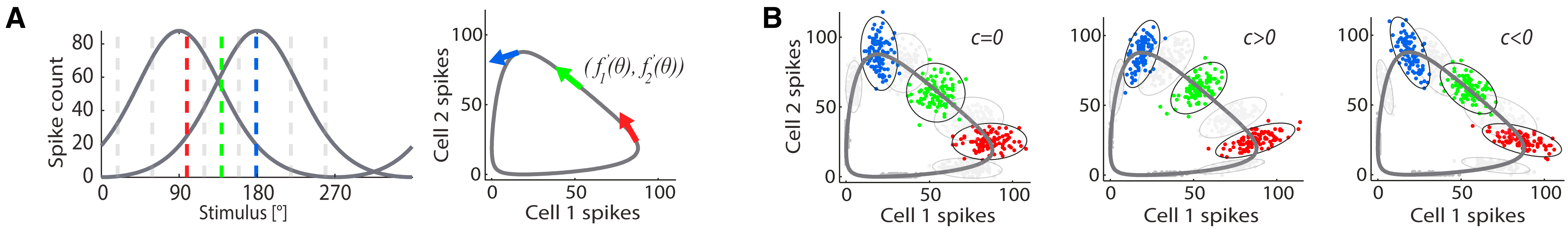
$$I_0 = \mathbf{m}'(s)^T \mathbf{\Sigma}^{-1} \mathbf{m}'(s) \sim O(N)$$

$$\mathbf{\Sigma}' = \mathbf{\Sigma} + \epsilon \mathbf{m}'(s) \mathbf{m}'(s)^T$$

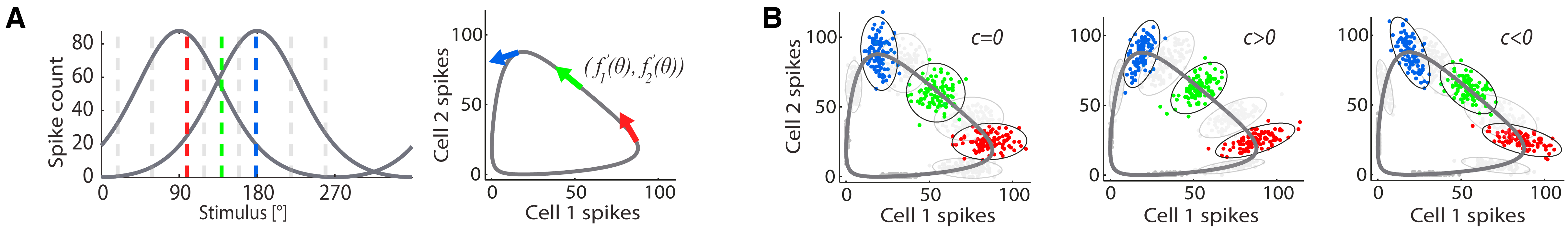
$$I' = \frac{I_0}{1 + \epsilon I_0}$$

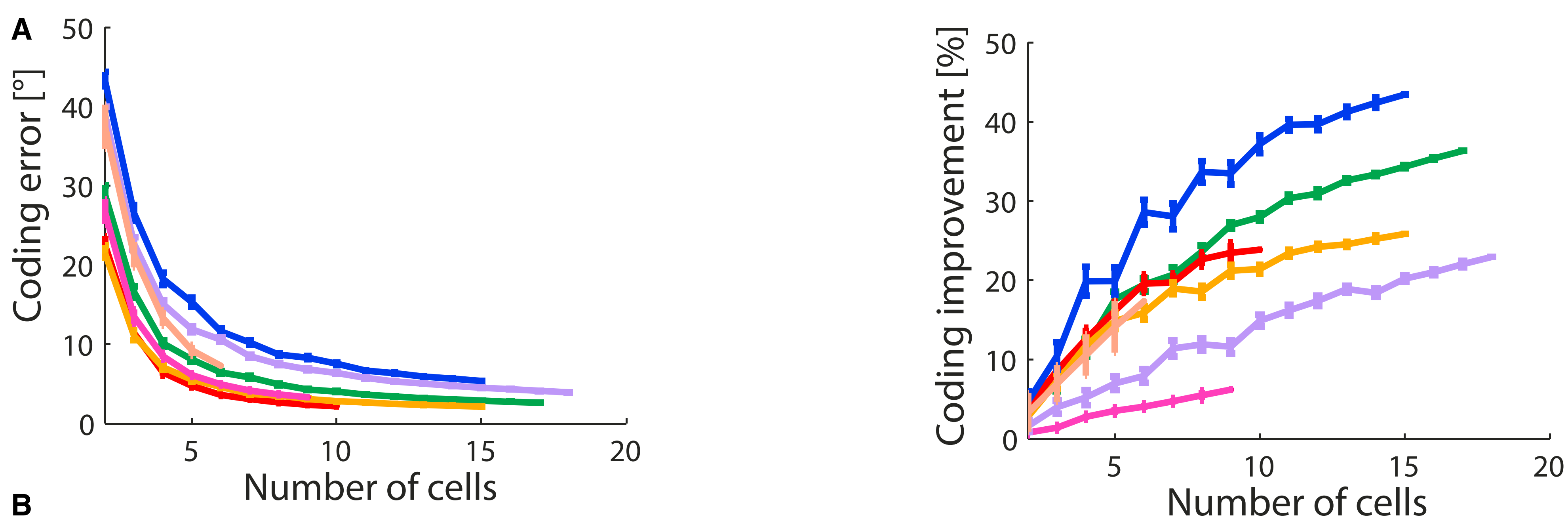


Noise Correlation can benefit information coding



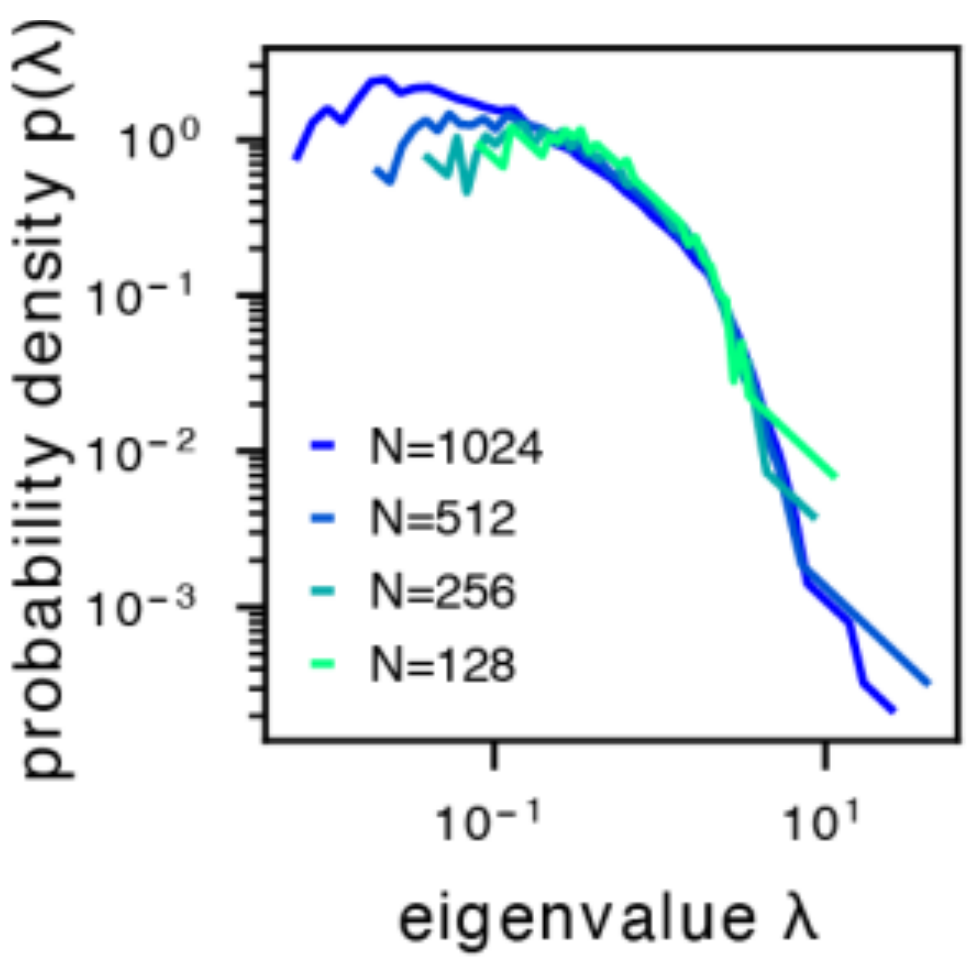
Noise Correlation can benefit information coding

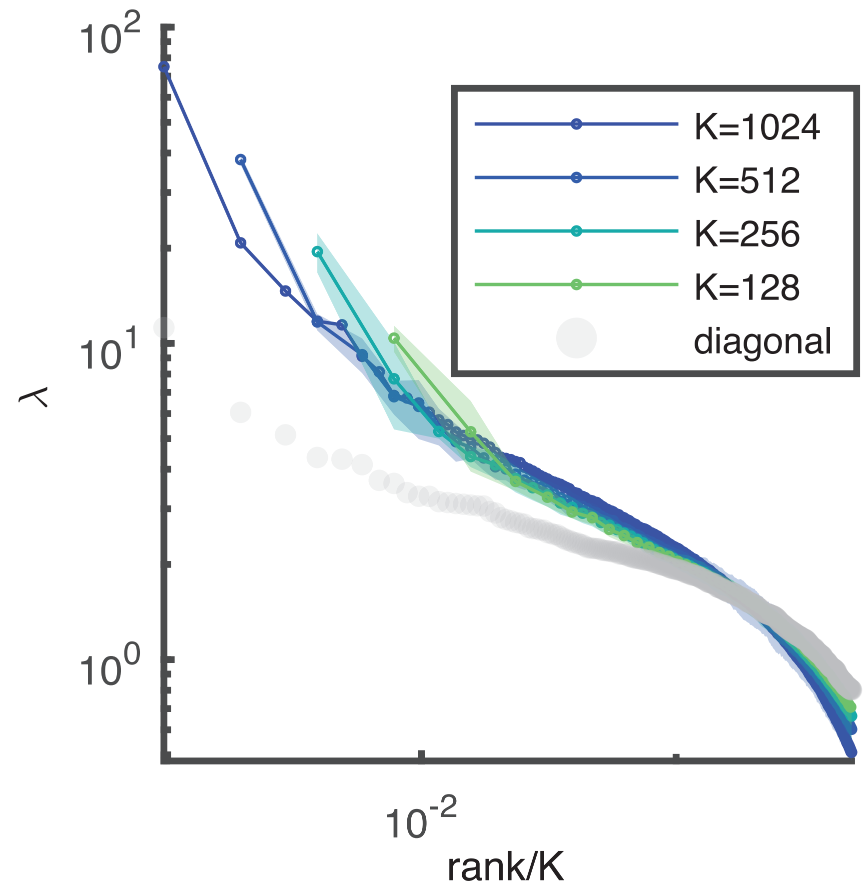




Franke et al, 2016

The nearly collapse of covariance eigenspectrm under random subsampling





Zezhen Wang et. al., BioRxiv, 2023

$$\lambda \sim (k/N)^{-\alpha}, \text{ with } \alpha < 1$$

Article

High-precision coding in visual cortex

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SUMMARY

Individual neurons in visual cortex provide the brain with unreliable estimates of visual features. It is not known whether the single-neuron variability is correlated across large neural populations, thus impairing the global encoding of stimuli. We recorded simultaneously from up to 50,000 neurons in mouse primary visual cortex (V1) and in higher order visual areas and measured stimulus discrimination thresholds of 0.35° and 0.37°, respectively, in an orientation decoding task. These neural thresholds were almost 100 times smaller than the behavioral discrimination thresholds reported in mice. This discrepancy could not be explained by stimulus properties or arousal states. Furthermore, behavioral variability during a sensory discrimination task could not be explained by neural variability in V1. Instead, behavior-related neural activity arose dynamically across a network of non-sensory brain areas. These results imply that perceptual discrimination in mice is limited by downstream decoders, not by neural noise in sensory representations.